THE T7 SYSTEM

The T7 system in its current – and – past – 2014 version – is inherently unfair in the sense that clients at a higher level in the hierarchy are allowed facilities which are orders of magnitude larger than those at the lowest level. From the point of mobile app users – the system amounts to criminal breach of trust. The unbelievable fact is that not a single client has raised this inherent unfairness issue at the government level of the Ministry of Finance. And this rot – is deployed across most major exchanges globally Are SEBI and the MoF aware of the issues - and what is their opinion in this matter - there has been no concealment of facts directly - but the common investor - one of the 40 million in India - lacks the technical capability to understand that this is a criminal breach of trust

Sagar Kapadia

Tuesday, 21 April 2026

Architectural Issues

T7 System – Architectural Issues

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T7 System – Architectural Issues

1. The Tired Latency Architecture is inherently unfair to non-co-located clients

Assertion:

- The Tiered Latency Architecture – is inherently unfair from the point of TBT/Market data

Explanation:

The Hierarchy

- Co-Loc Servers for Brokers who can afford the shelf space and the FPGA price tag
- Fiber Optical Cables for brokers who cannot afford to co-Locate
- Sub-Brokers who connect to brokers - via Leased Line
- Apps like Zerodha for retail investors connecting over 5G

Co-Loc Servers

- The co-located servers get TBT data in virtually real time due to Fiber optic cable length of a few meters
- They can respond in nano-secs due to FPGA architecture

Remote Brokers on standard software/hardware systems

- The remote brokers who are connected via longer length Fiber optic cable or leased lines get TBT data at order-of-magnitude time lag compared to co-loc brokers
- Their response time is also much greater - because they use hardware/software system - not purely hardware systems

Sub-brokers

- Leased Lines connect their systems to the brokers both co-loc / standard
- Orders of magnitude slower data

Retail Investors via App on 5G

- Slowest receipts - stale data which is actually dangerous - they see a fabricated reality

T7 System – Architectural Issues

- Most of the thousands of TBT packets broadcast by the exchange are simply filtered out/dropped - due to lack of bandwidth

Implications

Receiving Tick-by-Tick (TBT) information earlier than others in financial exchanges provides a significant competitive advantage by enabling latency reduction, high-frequency trading (HFT) superiority, and superior market liquidity assessment. TBT data transmits every single transaction and order modification in real-time, unlike conventional snapshot data which acts in time-based intervals. [1, 2, 3, 4]

- Latency Advantage (First-to-Market): In exchanges using a TBT protocol, information is often transmitted in the sequence of orders received. Getting earlier access to this data flow—often facilitated by co-location (placing servers close to the exchange)—means traders can react to price movements or order book changes milliseconds or even microseconds faster than others.
- Superior Execution for Scalpers and HFTs: High-frequency traders and scalpers, who rely on small price fluctuations, can use the early data to fill their orders at more favourable prices before the market moves. This speed advantage can directly translate into higher profits, as they can execute, cancel, or modify orders faster than the general market.
- Order Book Transparency and Depth: Early access allows for a more detailed view of the entire exchange order book, revealing market depth beyond the best five bids/asks. This reveals the hidden volume of larger players (institutional investors) who are planning to buy or sell massive quantities, allowing early recipients to adapt their strategies ahead of large price shifts.
- Better Risk Management: With access to the complete, real-time data stream, traders can place more accurate stop-loss orders, reducing slippage during periods of high volatility.
- Improved Decision-Making: Early TBT data filters out the low-activity period, reducing "noise" and providing a clearer picture of actual market movements. This enhances the accuracy of technical indicators, allowing for better identification of breakouts and trends. [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
- Essentially, receiving TBT information earlier allows market participants to "front-run" the information flow, turning a split-second advantage into consistent trading profitability. [2, 9]

T7 System – Architectural Issues

Evidence

- Order Entry
- Co-Location topology
- Order Entry Network
- Market Data Network
- PS vs LF Type Gateways
- Order Entry Latency – Comparison of Access Types
- Recommendations for Speed

2. Algorithmic Trading is Inherently unfair to human decision makers

Assertion

Pitting a biological – Human - Brain against a Silicon Brain – A Computer - CPU/Software/OS or an FPGA – where the algorithm itself is configured on to specialized/bespoke system - is inherently unfair

Explanation

The Silicon brain is way faster than a human brain

If the silicon brain is an FPGA system – the disadvantages are orders of magnitude even higher than the CPU/Software/OS silicon brain

Implications

The response time of a biological brain is thousands or millions of times lower than that of the silicon brain. This implies that the algorithmic server – can execute thousands of transactions by the time the human brain finishes with one transaction. The extremely brief latency gap between transactions with the silicon brain allow it to make full use of the PS bandwidth and the Matching engine's throughput

The debate over whether algorithmic trading (also known as "algo" or "HFT" - High-Frequency Trading) is inherently unfair to human investors is a central theme in modern financial regulation. From a technical and structural standpoint, the "unfairness" usually stems from three distinct advantages that machines have over humans: **Time, Access, and Information Integrity.**

T7 System – Architectural Issues

1. The Physical Advantage: Co-location and Fiber

Human investors typically connect to the market via a retail app or a web browser. This journey involves several "hops": from your phone to a cell tower, across the public internet, into a broker's server, and finally to the exchange.

Algorithmic firms pay for **Co-location**, placing their servers in the same room as the exchange's matching engine.

- **The Difference:** While a human sees a price update in 50 to 500 milliseconds, an algo firm sees it in microseconds (1/1,000,000th of a second).
- **The Result:** By the time a human clicks "Buy," the algorithm has already seen the order coming, bought the available shares at a lower price, and sold them back to the human at a fractionally higher price.

2. Protocol Differences: UDP vs. TCP

As you have identified in your research, the technical protocols used to send data differ between professionals and retail investors.

- **Retail (TCP):** Most retail apps use TCP, which is "reliable"—it ensures every packet of data arrives in the correct order. However, checking for errors takes time.
- **Professional (UDP):** High-frequency feeds like **EOBI** (Enhanced Order Book Interface) often use UDP. This protocol is "fire and forget." It doesn't wait for a confirmation that the data was received; it just blasts it out as fast as possible.
- **The Unfairness:** Professional algos prioritize **speed over accuracy**. They would rather lose a single packet of data than be 10 microseconds late. Humans, stuck on slower "reliable" connections, are essentially watching a "delayed broadcast" of a game that has already moved on.

3. The "Global Reality" vs. Sharded Partitions

Modern exchanges are "sharded" into multiple partitions to handle millions of orders. Each partition (a group of stocks) operates its own matching engine.

- **The Issue:** There is often no "Global Master Clock" that synchronizes every single partition at the nanosecond level.
- **The Result:** A sophisticated algorithm can "triangulate" what is happening across different partitions faster than the exchange can

consolidate that data for the public. This allows algos to predict price movements in one stock based on a trade that just happened in a related stock on a different partition. The human investor sees these as two separate, unrelated events.

4. The "Public Data First" Gap

In many modern systems (like the T7 architecture), the system is designed to broadcast the public market update (the "Tick") slightly *before* it sends the private confirmation to the person who actually made the trade.

- **The "Predatory Window":** This means if you place a large order, an HFT algorithm might see your order executed on the public feed and react to it before your own broker's app even tells you that your order was filled.

Is it a "Criminal Breach of Trust"?

From a regulatory perspective (SEBI/MoF), these features are often defended as necessary for **Market Liquidity**. The argument is that algorithms provide the "buy" and "sell" orders that allow humans to trade instantly.

However, the "criminal breach of trust" argument arises when one considers that the **40 million retail investors** are participating in a game where the physical and digital rules are tilted in favor of those who can afford the "entry fee" for co-location and FPGA hardware. While the facts are not hidden, they are buried in technical manuals that require significant expertise to decode, creating a massive gap between "transparency" and "actual understanding" for the common citizen.

Evidence

- Recommendations for Speed
-

3. No Global Clock (The "Fabricated Reality")

Assertion:

- Each partition is consistent, but the global system has no "True Now."

T7 System – Architectural Issues

Explanation

Timestamps and sequences are local to the Partition Matching Engine. There is no global clock or sequencer

Implications

If a client is looking at a "Market Watch" with stocks from Partition 1 (Reliance) and Partition 2 (HDFC), there is no single sequence number that tells them which trade happened first across the two partitions. They are seeing two different "clocks" merged into one screen.

Evidence

- Partitions

4. Mobile Apps get a highly truncated stream of market data with large [orders of magnitude] latency as compared to co-located clients

Assertion:

Non-Co-located clients and Mobile Apps get a highly truncated stream of market data depending upon their level

Explanation

The matching engine is capable of 1,50,000 transactions per second – corresponding to the transactions, an equivalent number of market data packets are generated. At max, mobile apps can handle 5-6 packets per second. The other packets are simply dropped

Implications

The view of the market from the mobile app - is a faded black and white photograph – of a non-existent reality – whereas the co-located brokers see a view in 8k – they receive all packets broadcast by the engine – in real time – and – it is important to note since there is no global state of the system – each broker may be experiencing a different hallucinated view of something that does not really exist

T7 System – Architectural Issues

Evidence

To do

5. The "Public Data First" Gap

Assertion

The system itself is architecturally designed to prioritize the public feed (EOBI) over the private confirmation (ETI).

Explanation

The T7 system sends out the UDP market data update *before* it sends the TCP confirmation back to the person who actually placed the order.

Implications

This creates a window (often several microseconds) where high-frequency traders in co-location can see your trade executed on the public feed and react to it before you even receive your own "Trade Confirmed" message. The system technically "tips off" the fastest players.

Evidence

- Latency Characteristics of EOBI vs ETI for Futures

6. UDP vs. TCP: High Speed/low latency preferred over Correctness [Guaranteed In-Order Delivery]

Assertion

- Using UDP (EOBI) is "irresponsible" because it sacrifices guaranteed delivery for speed.

Explanation

Prioritizing speed over delivery guarantees may result in dropped packets, out of sequence packets at all levels – including the co-located clients. – It tries to re-invent the wheel – and ends up making a square wheel – which it then coerces its clients to use for their Ferrari

T7 System – Architectural Issues

[Co-Located Servers] to bicycle [Mobile Apps]. The responsibility of data integrity has been off-loaded to the clients – which only the co-loc level brokers have the financial capability to attempt – again – without any guarantee – the others - put their faith in God

Implications

There are layers / sub-systems which attempt to replicate the TCP guarantee of packets being received and received in-order – at the Application Level of the Network Stack - but the T7 system makes no guarantees – and cannot make - absolute guarantees due to UDP protocol rather than the TCP protocol

UDP causes Jitter in Market Data

Evidence

- Timestamps - Description
- Timestamps – Reference
- Latency Characteristics of ETI vs EOBI vs EMDI
- EOBI Latency of Primary and Secondary Feed
- Retransmission Service over TCP

T7 System – Architectural Issues

1. The Documentary Evidence

Speed/Latency Hierarchy

Order Entry

Order Entry

Co-Location Network

Participants may use Co-Location to place their infrastructure in the datacentre that hosts the T7 system.

The Co-Location 10 Gbit network has the following properties:

Fair and equal access

Regardless of the Co-Location room we ensure all lines are created equal.

More precisely the latency between the handover point in the participant's rack and the first (Access Layer) switch is calibrated to below ±1.75 ns. The reduction from ±2.5 ns was achieved as an intermediate step and Deutsche Börse plans to further improve the cable length normalization.

Two redundant halves ('A' and 'B')

There are two independent order entry network halves.

As active gateways are placed in either half there is an optimal side for each gateway (even numbered gateways are on the A side, odd number on the B side). The only exception is FIX LF: There is only one active FIX LF gateway which is by default located on the B-side.

Crossing sides, e.g. connecting to a B side gateway via an A network, is possible but results in about 1 µs higher base latency.

Two hierarchical switch layers

Participants connect to Access Layer switches (currently 9 Eurex*, 2 Xetra per side).

The uplink of each Access Layer switch is connected to a Distribution Layer switch.

The Distribution Layer switches have a direct connection to the active gateways on the respective side.

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* For capacity reasons two Access Layer switches were added to each side for Eurex on 22 February 2025.

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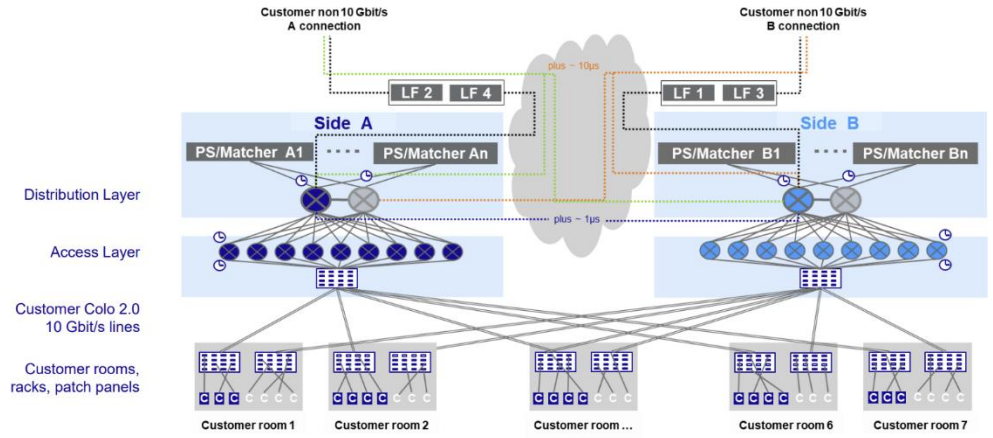
Co-Location Topology

Network Topology in Co-Location

Eurex Order Entry



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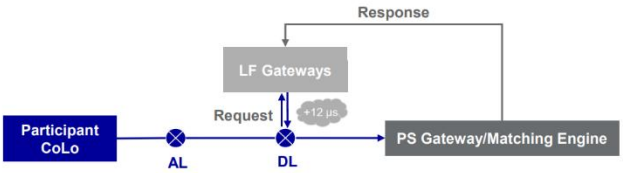
Gateway Types [PS and LF]

PS Gateway versus LF Gateways

Latency comparison



- Using 10 Gbit/s cross connects and access via PS gateways provides the fastest way for order and quote management in T7.
- LF gateways on the other hand allow access to all partitions of a market via a single session.
- Some markets (e.g. XMAL) are using LF gateways only.
- Most markets have combined PS gateway/ Matching Engines in place for which all Matching Engine bound requests sent to LF gateways are routed via PS gateways.
- The base inbound latency for LF gateway transactions is around 12 µs higher compared to directly accessing the PS gateways. Note also that requests that have to cross sides between LF and PS gateways take another ~1 µs to reach the Matching Engine. This was significantly reduced from ~45 µs to ~1 µs end of January 2025.

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Order Entry Network

Order Entry

Introduction

Requests sent to T7 will be routed via an access network and a gateway.

There are the following basic connection alternatives:

Choice of Network

There are two network classes connecting a participant's installation with the T7 gateways:

- Co-Location with 10 Gbit bandwidth and a one-way base latency of around 2 µs.
- Other networks with less than 1Gbit bandwidth and a one-way base latency of minimum about 50 µs.

Choice of Session Type

T7 supports three session types:

- High frequency sessions connect to PS gateways for low latency access to a single partition (flat binary protocol: ETI). Please note that HF sessions can only be used within Equinix FR2 data center.
- Low frequency sessions connect to LF gateways for convenient access to all partitions, with a considerably higher base latency (ETI).
- FIX LF sessions connect to FIX LF gateway for convenient access to all partitions using the FIX protocol, with a considerably higher base latency than LF gateways.

Remarks

- LF gateway and FIX LF requests are routed via PS gateways.

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T7 System – Architectural Issues

Market Data Networks

Market Data

Overview

Market Data can be consumed over two distinct types of networks and in various types

Choice of Network

There are two network classes available for market data:

- Co-Location with 10 Gbit bandwidth and a one-way base latency of around 2 µs.
  10 Gbit connections are equalized in length (cable latency difference of less than ±1.75 ns) and provide the lowest jitter.
  The reduction from ± 2.5 ns was achieved as an intermediate step and Deutsche Börse plans to further improve the cable length normalization.
- Other networks with less than 1Gbit bandwidth with higher base latency.

Choice of Market Data Type

There are three market data types:

- Order by Order market data (EOBI) with highest granularity and lowest latency in flat binary format.
  EOBI is sent out directly from the Matching Engine and is only available via 10 Gbit network.
- Price level aggregated market data (EMDI) with slightly higher latency in FAST encoded format.
- Netted price level aggregated market data (MDI) in FAST encoded format.

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Order Entry Latency – Comparison of Access Types

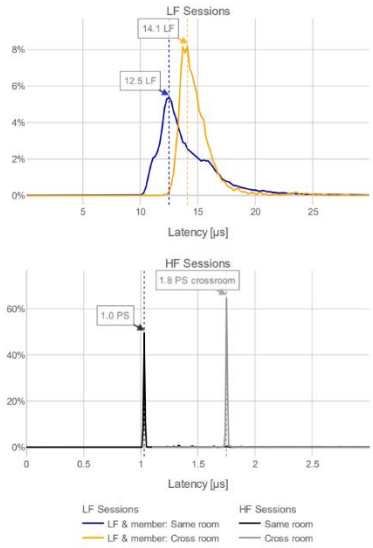
Order Entry Latency

Comparison of access types

The table below shows a comparison of different access options to the T7 system. All times given are in microseconds.

Network timestamps (t_3a) are synchronized using white rabbit. The time synch quality between these timestamps is thus ~1 ns. Other T7 timestamps are subject to jitter of up to ±50 ns.

Gateway type	Same side line	Same side partition	t_3a to t_3n uncongested latency
PS	☑	☑	1
	☐	☑	2
LF	☑	☑	12
	☑	☐	13
	☐	☑	13
	☐	☐	14



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Recommendations for Speed

What you need to be fast…

A few recommendations to achieve the low latency

Use the Equinix Co-Location facility to be close to Deutsche Börse T7.

Use state-of-the-art switches (if any) and only have at most one hop between the exchange network and your server. Alternatively, use hardware solutions to connect directly without hops (e.g. FPGA).

Use good network interface cards and TCP/IP acceleration, e.g. a kernel-by-pass library.

Use at least two 10 Gbit/s (cross-) connections in Co-Location for EOBI market data and two 10 Gbit/s connections for T7 ETI.

Use HF sessions to connect to PS gateways and make sure you use the cross connect on the same side as the gateway you are connecting to (compare time-to-live values in the IP header in the responses from both sides).

Measure and analyze your own timestamps (e.g. the reaction time as recommended on the next slide).

Use state of the art time synchronization, i.e. GPS clocks and a high-quality time distribution. The PTP signal you can get from us has a quality of ±50 ns. For our network timestamps we use the White Rabbit protocol and PPS breakouts. You can connect to our white rabbit time service providing you a time synchronization quality of 1-2 ns max, see https://www.deutsche-boerse.com/dbg-en/products-services/ps-technology/ps-connectivity-services/ps-connectivity-services-time-services.

We provide highly accurate network timestamps of all orders leading to a market data update via the high precision timestamp file service, see https://www.mds.deutsche-boerse.com/mds-en/analytics/high-precision-timestamps.

Use the EOBI Execution Summary for fast trading decisions and position keeping (passive executions). For a consistent order book, all incremental updates following the Execution Summary should always be processed. For fastest decisions evaluate the market data classification based on the DSCP flags in the IPv4 header of EOBI market data packets.

Trade notifications need to be processed to create safety. We recommend to use either a low-frequency ETI session or a FIX trade capture drop copy to confirm the fast execution information provided by the execution reports via high-frequency sessions.

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2. Algorithmic / HFT Trading Inherently Unfair

Recommendations

What you need to be fast…

A few recommendations to achieve the low latency

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Use good network interface cards and TCP/IP acceleration, e.g. a kernel-by-pass library.

Use at least two 10 Gbit/s (cross-) connections in Co-Location for EOBI market data and two 10 Gbit/s connections for T7 ETI.

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T7 System – Architectural Issues

Order Entry - Gateways

Order Entry
Gateways

There are three gateway types to access the T7 system:

Partition-specific (PS) Gateway combined with Matching Engine

Protocol:	flat binary (ETI)
Allowed session types:	High Frequency Sessions only
Sequencing:	FIFO operation (Sequence guaranteed from network card to Matching Engine in)
Latency:	lowest, median latency ~ 12 µs network card to Matching Engine in
Versatility:	Allows routing to one partition only, only subset of broadcasts available

Low Frequency (LF) Gateway

Protocol:	flat binary (ETI)
Allowed session types:	Low Frequency Sessions only
Sequencing:	FIFO not guaranteed
Latency:	medium, (additional ~32 µs latency compared to PS gateway direct access)
Versatility:	Routes to all partitions (via PS gateway), all ETI broadcast types available

FIX LF Gateway

Protocol:	FIX
Allowed session types:	Fix Sessions only
Sequencing:	FIFO guaranteed
Latency:	high, requests to the Matching Engine are routed via PS gateways
Versatility:	Routes to all partitions (via PS gateway), all FIX broadcast types available

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Order Entry - Introduction

Order Entry
Introduction

Requests sent to T7 will be routed via an access network and a gateway.

There are the following basic connection alternatives:

Choice of Network

There are two network classes connecting a participant's installation with the T7 gateways:

- Co-Location with 10 Gbit bandwidth and a one-way base latency of around 2 µs.
- Other networks with less than 1Gbit bandwidth and a one-way base latency of minimum about 50 µs.

Choice of Session Type

T7 supports three session types:

- High frequency sessions connect to PS gateways for low latency access to a single partition (flat binary protocol: ETI). Please note that HF sessions can only be used within Equinix FR2 data center.
- Low frequency sessions connect to LF gateways for convenient access to all partitions, with a considerably higher base latency (ETI).
- FIX LF sessions connect to FIX LF gateway for convenient access to all partitions using the FIX protocol, with a considerably higher base latency than LF gateways.

Remarks

- LF gateway and FIX LF requests are routed via PS gateways.

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T7 System – Architectural Issues

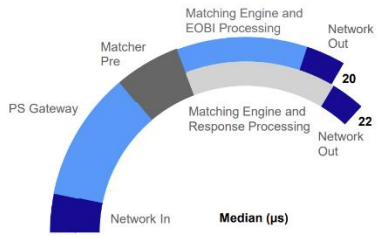
Latency Composition

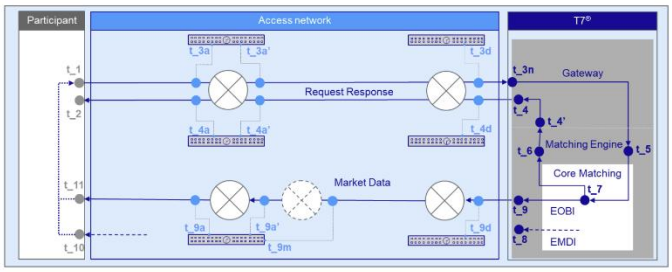
T7® Latency

Composition

The T7 trading system provides utmost transparency about its latency characteristics. Most of the timestamps taken at the various measurement points within T7 Core are included in each ETI response and EOBI market data. With the high precision timestamp file, we also make three network timestamps available for each EOBI market data packet (t_9d) and it's triggering transaction (t_3a, t_3d).

The latency circle shows the median latencies for the request-response/EOBI market data path for Eurex futures sent via high frequency sessions measured on 13 February 2025.





3. UDP vs TCP - Speed over Correctness

Timestamps - Description

T7® Timestamps

Description

t_[x]a, t_[x]a'	time taken by network capture devices in the Access Layers.
t_[x]d	time taken by network capture devices in the Distribution Layers.
t_9m	time taken by network capture device between Access Layer and Mid-layer.
t_1, t_2	can be taken by a participant (e.g. via a network capture) when a request/ response is read from/written to the network.
t_3n	time taken by the PS gateway when the first bit of a request arrives on the PS gateway NIC; contained in (private) ETI response for PS gateway enabled partitions. Consecutive messages via the same session may be assigned to the same t_3n.
t_3	time taken by the ETI gateway application when a request is read from the socket on the participant´s side of the gateway; contained in (private) ETI response for transactions for non-PS gateway enabled partitions (e.g. XVIE).
t_4'	time taken by the ETI gateway when a response/ notification is received by the ETI gateway from the Matching Engine; contained in (private) ETI response/ notification.
t_4	time taken by the ETI gateway when a response/ notification is written to the socket on the participant´s side of the gateway; contained in (private) ETI response/ notification.
t_5, t_6	time taken by the Matching Engine when a request/response is read/written; contained in (private) ETI response.
t_7	time at which the Matching Engine starts maintaining the order book
t_8	time taken by EMDI publisher just before the first respective UDP datagram is written to the UDP socket.
t_9	time taken by EOBI publisher just before the first respective UDP datagram is written to the UDP socket.
t_10, t_11	can be taken by a participant (e.g. via a network capture) when a UDP datagram is read from the UDP socket.

T7 System – Architectural Issues

Timestamps – Reference

T7® Timestamp Reference

The timestamps t_3,…,t_9 are available via the following fields:

Timestamp	Tag no.	Field name	Present in
t_3, t_3n	5979	RequestTime	ETI Response EMDI Depth Incremental message, in case a trade is reported EOBI Execution Summary, Order Add, Order Modify, Order Modify Same Priority and Order Delete messages
t_4'	7765 25043	ResponseIn NotificationIn	ETI Response (from the Matching Engine) ETI Notification (from the Matching Engine)
t_4	52	SendingTime	ETI Response and Notification
t_5	21002 2445	TrdRegTSTimeIn AggressorTime	ETI Response (from the Matching Engine) EMDI Depth Incremental message, in case a trade is reported EOBI Execution Summary message
t_6	21003	TrdRegTSTimeOut	ETI Response and Notification (from the Matching Engine)
t_7	17 273 60 21008	ExecID MDEntryTime TransactTime TrdRegTSTimePriority	ETI Response (from the Matching Engine) EOBI Execution Summary message EMDI Depth Incremental, Depth Snapshot and Top of Book Implied message EMDI messages for other events EOBI Order Modify Same Priority and Order Delete messages EOBI Order Add and Order Modify messages
t_8	No tag	SendingTime	T7 EMDI UDP packet header
t_9	60	TransactTime	EOBI packet header
t_8 - t_5	No tag	PerformanceIndicator	EMDI UDP packet header of the T7 EMDI Depth Incremental stream

Notes on timestamps:
All timestamps provided are 8 byte integers (in nanoseconds after Unix epoch).
The PerformanceIndicator is a 4 byte integer (in nanoseconds).
The Network timestamps (t_[x]a, t_[x]a', and t_[x]d, t_[x]d') are not available in any protocol field but some via the High Precision Timestamp File service.

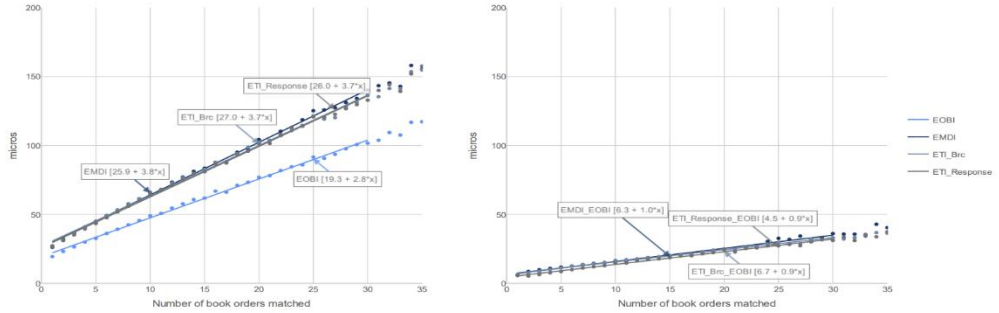
Latency Characteristics of ETI vs EOBI vs EMDI

Trading System Dynamics

Latency characteristics of ETI versus EOBI versus EMDI

The diagrams below display the dependency of the median latency on the complexity of a trade for ETI (the right diagram is normalized to EOBI) (t_4 - t_7), EMDI (t_8 - t_7) and EOBI (t_9 - t_7). Note that for ETI we display the gateway sending time of the first passive notification and for EOBI the sending time of the UDP datagram containing the Execution Summary message.

The difference between public and private data has slightly been narrowed with the integration of EMDI into Matching Engine process. However, the 'public data first' principle is still being ensured. In over 99% of all trades, we disseminate order book data on EOBI first (also for larger trades).



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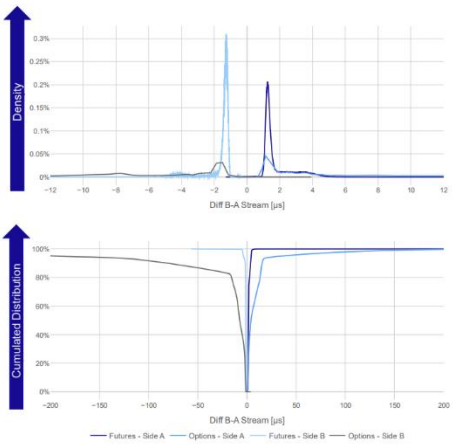
T7 System – Architectural Issues

EOBI Latency of Primary and Secondary Feed

Trading System Dynamics

EOBI latency difference of primary and secondary feed



For products assigned to even partitions, market data is published first on the A and then on the B stream. For products assigned to odd partitions market data is published first on the B and then on the A stream. This also applies to a failover scenario.
The partition ID / product ID is contained in the UDP datagram header of the order book incremental messages and can be used for filtering on UDP datagram level for EMDI / EOBI.
Furthermore, a UDP datagram on the T7 EMDI / EOBI order book delta or snapshot channel contains data of exactly one product (e.g. EURO STOXX 50® Index Futures).

The data for both primary and secondary streams is disseminated by the same server using two ports connected to the two sides of the network. The process sends the data first on the primary interface. After all datagrams of a transaction are sent it starts sending on the secondary interface.

The median latency difference between the A and the B EOBI incremental feed is about 1.7 µs for futures and Cash products. For options the median is slightly higher and there are far more outliers (i.e. much slower secondary feed). The reason is that since data is published on the secondary feed only after all datagrams of a transaction are sent on the primary feed, the latency difference depends on the complexity of the transaction, i.e. a mass quote with 200 quote updates will lead to a higher delay than a single order entry.

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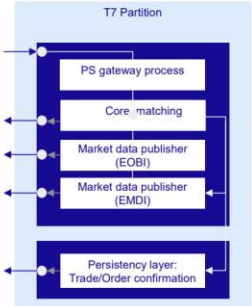
4. No Global State of the Market

Partitions

T7® Topology

Partitions



Orders/quotes entered for a specific product are sent by the PS gateway process to the core Matching Engine process (both residing on the same server in the same partition).

The matching priority is assigned when the orders/quotes are read into the Matching Engine.

The core matching component works as follows:
- when an order/quote arrives, it is functionally processed (e.g. put in the book or matched),
- handover of data to the EOBI market data publisher, followed by EMDI market data publisher and
- handover of all data resulting from the (atomic) processing of the incoming order/quote to persistency component in the partition.

Resulting responses and private broadcasts are sent out in the following order:
- direct response to the order/quote entered (for persistent as well as for non-persistent orders and quotes),
- fast execution information for booked orders/quotes (in case of a match).

The generation of listener broadcasts, trade confirmations (by the persistency layer) and of non-EOBI/non-EMDI market data is done by separate processes. Hence the order of the resulting messages from these processes is not strictly deterministic.

Note that the Matching Engine holds states of orders in memory. All responses, broadcasts, EOBI and EMDI market data thus are preliminary by nature.

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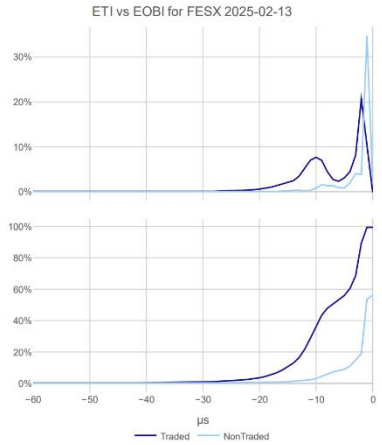
5.The "Public Data First" Gap

Latency Characteristics of EOBI vs ETI for Futures

Trading System Dynamics

Latency characteristics of EOBI versus ETI for Futures



T7 is designed to publish order book updates first on its public data feed.

The diagram shows the time difference distribution between public and private data in microseconds (EOBI first datagram vs ETI responses, t_9 - t_4).

The data is a production sample from 13 February 2025.

EOBI market data is in median 1.5 µs faster than the ETI response for order book updates and 7.5 µs faster for executions.

The first EOBI datagram is faster in more than 99.4 percent of the cases compared to the ETI response, and also the first passive ETI book order notification for simple transactions.

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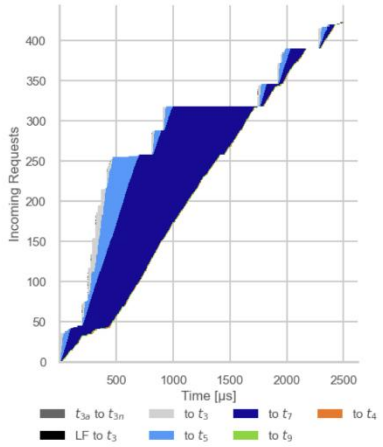
T7 System – Architectural Issues

Bursts/Spikes

Eurex Micro-Burst Dynamics

T7® Latency

Eurex Micro-burst dynamics



During micro-bursts, the input into the trading system may be greater than the throughput capabilities. This in turn causes queuing which results in higher latencies.

Higher latencies cause risk (i.e. it takes longer to place/pull an order).

T7 provides real-time performance insights by providing up to six timestamps with each response and key timestamps with every market data update.

The left chart shows for one example burst in FESX on 13 February 2025 the following paths:

- Network Access Layer (t_3a) to PS GWIn (t_3n) to PS GW SWIn (t_3) to
- Matching Engine in (t_5) to
- Start matching (t_7) to
- EOBI SendingTime (t_9) [where available] to
- ETI SendingTime (t_4).

Typical throughput rates in kHz (1/ms) are 8000 at t_3n, ~500 at t_5, ~300 at t_7.

EOBI send times are usually before the gateway send time of responses.

Note that base latency for requests entered via LF gateways is ~ 12 µs higher. As all requests are routed via PS gateways no overtaking between LF gateway and PS gateway requests is observed.

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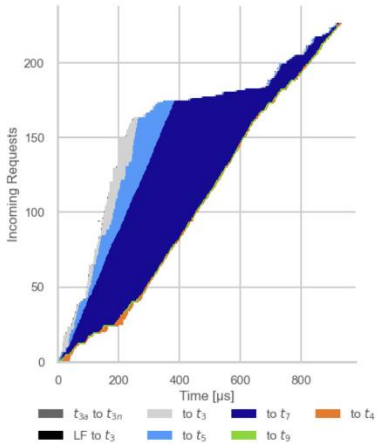
Xetra Micro-Burst Dynamics

T7® Latency

Xetra Micro-burst dynamics



During micro-bursts, the input into the trading system may be greater than the throughput capabilities. This in turn causes queuing which results in higher latencies.

Higher latencies cause risk (i.e. it takes longer to place/pull an order).

T7 provides real-time performance insights by providing up to six timestamps with each response and key timestamps with every market data update.

The left chart shows for one example burst on Xetra partition 57 (4 June 2024) the paths:

- Network Access Layer (t_3a) to PS GWIn (t_3n) to PS GW SWIn (t_3) to
- Matching Engine in (t_5) to
- Start matching (t_7) to
- EOBI SendingTime (t_9) [where available] to
- ETI SendingTime (t_4).

Typical throughput rates in kHz (1/ms) are 8000 at t_3n, ~500 at t_5, ~300 at t_7.

EOBI send times are usually before the gateway send time of responses.

Note that base latency for requests entered via LF gateways is ~ 12 µs higher. As all requests are routed via PS gateways no overtaking between LF gateway and PS gateway requests is observed.

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6. References

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